

Introduction to Computer Simulation

Home Work 10

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Question 2

2.a)

We can easily verify this with some simple code:

```
import numpy as np
import matplotlib.pyplot as plt

N = 100

X_i = np.random.rand(N)

X_i_sqrd = X_i**2

X_i_sqrd_mean = X_i_sqrd.mean()

X_i_mean = X_i.mean()

sigma_x_sqrd_1 = X_i_sqrd_mean - X_i_mean**2

indx = [0,1,2,3,4,5,6,7,8,9,10]

list_x = np.zeros(len(indx))

for i in range(len(indx)):
    list_x[i] = (X_i**i).mean()
```

This leads us to the results:

$$\langle X_i \rangle = 0.53 \quad , \quad \langle X_i^2 \rangle = 0.36 \quad , \quad \sigma_{X_i}^2 = 0.087$$

These being close to the theoretical results, 0.5, 0.33 and 0.83. Further analyzing the behavior of $\langle X_i^n \rangle$ for distinct values of n we arrive at the graph:

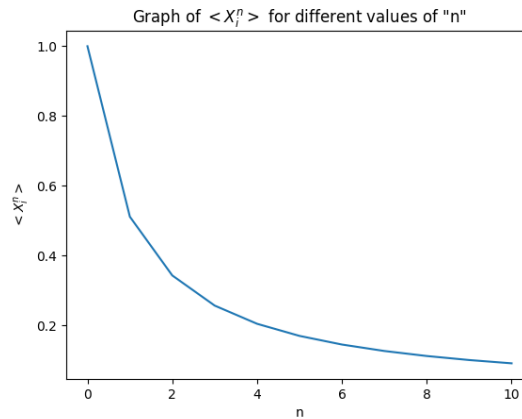


Figure 1: Q.2a

2.b)

The following code was used for task 2.b)

```
N = np.array([10,20,40])
m = int(1e5)

sigma_10 = np.zeros(m)
sigma_20 = np.zeros(m)
sigma_40 = np.zeros(m)

sigma = np.array([sigma_10,sigma_20,sigma_40])

for j in range(len(N)):
    for i in range(m):
        X_i = np.random.rand(N[j])
        sigma_x_sqrd = ((X_i-X_i.mean())**2).mean()
        sigma[j,i] = sigma_x_sqrd
```

2.c)

From question one we can calculate the theoretical value of $\langle \hat{\sigma}_{X_i}^2 \rangle$ for $N = 10, 20, 40$ and $\sigma_{X_1}^2 = 1/12$:

N	10	20	40
$\langle \hat{\sigma}_{X_i}^2 \rangle$	0.07500	0.07917	0.08125

Table 1: Expectation value of the variance estimator for different sample sizes N .

From the simulation we arrive at the values:

N	10	20	40
$\langle \hat{\sigma}_{X_i}^2 \rangle$	0.07500657	0.07916594	0.08123777

Table 2: Expectation value of the variance estimator for different sample sizes N .

Showing that the approximation:

$$\langle \hat{\sigma}_{X_i}^2 \rangle \approx \frac{1}{M} \sum_{j=1}^M \hat{\sigma}_{X_i}^{2(j)}$$

Holds well in the scale of $m = 10^5$

2.d)

Given that:

$$\langle \hat{\sigma}_{X_i}^2 \rangle = \sigma_{X_i}^2 \left(1 - \frac{1}{N} \right)$$

We can construct an estimator $\tilde{\sigma}_{X_i}$, such that $\langle \tilde{\sigma}_{X_i} \rangle = \sigma_{X_i}^2$, that is:

$$\tilde{\sigma}_{X_i}^2 = \sigma_{X_i}^2 \left(\frac{N-1}{N} \right) \quad (1)$$